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Chapter 1

Introduction to Operations Management

Operations Management
FOURTEENTH EDITION

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Chapter 1: Learning Objectives

You should be able to:

- LO 1.1 Define the terms *operations management* and *supply chain*
- LO 1.2 Identify similarities and differences between *production* and *service* operations
- LO 1.3 Explain the importance of learning about operations management
- LO 1.4 Identify three major *functional areas* of organizations
- LO 1.5 Summarize the two major aspects of *process management*
- LO 1.6 Describe the operations function and the nature of the *operations manager's job*
- LO 1.7 Explain the key aspects of *operations management decision making*
- LO 1.8 Briefly describe the *historical evolution* of operations management
- LO 1.9 Describe the current issues in business that impact operations management
- LO 1.10 Explain the key Issues for operations managers today
- LO 1.11 Explain the need to manage the supply chain

Introduction of Operations Management

What is operations?

- The part of a business organization that is responsible for producing goods or services

How can we define operations management (OM)?

- The management of systems or processes that create products and/or provide services

Good or Service?

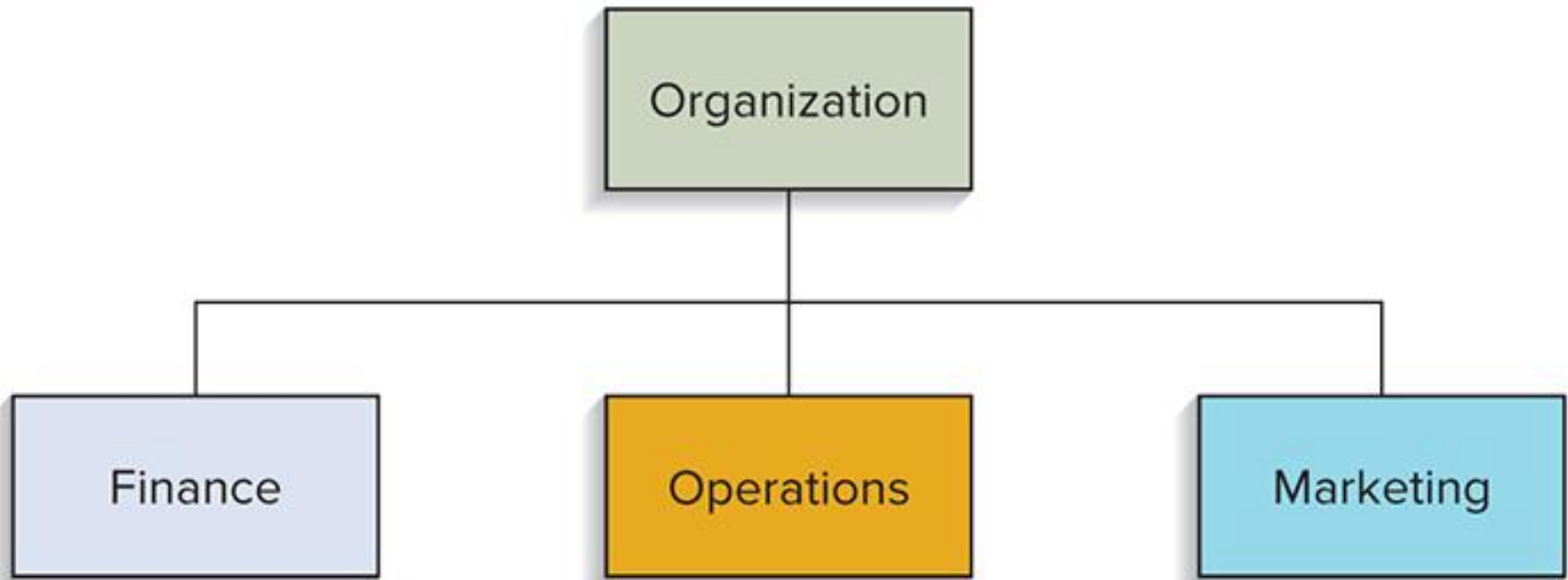
Goods (Products) are **physical items** that include raw materials, parts, components, and final products

- Automobile
- Computer
- Oven
- Shampoo

Services are **activities** that provide some combination of time, location, form or psychological value

- Air travel
- Education
- Haircut
- Legal counsel

Operations Functions of the Organization



Note: Three main **operations functional areas** of typical organizations

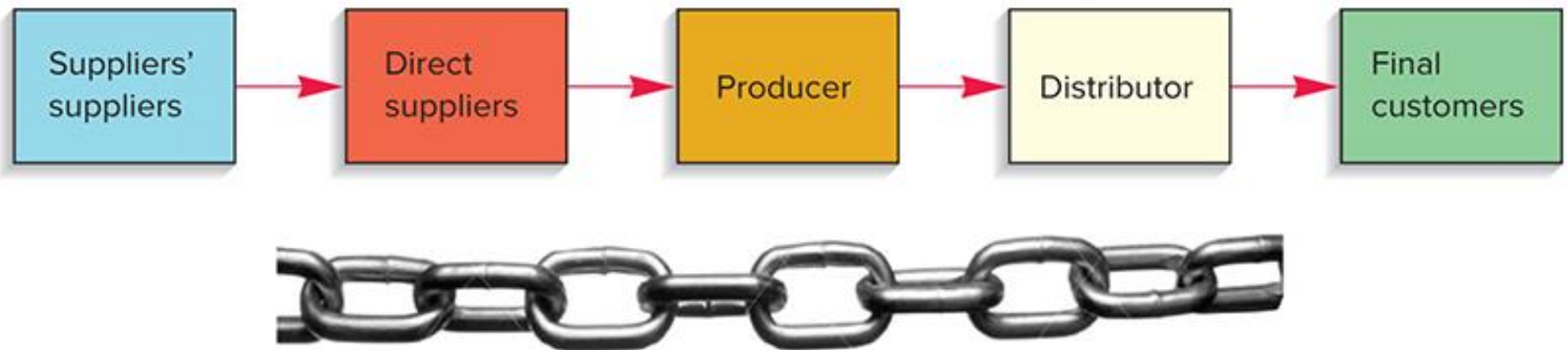
Operations Functions of the Organization

- **Finance** is responsible for securing financial resources at favorable prices and allocating those resources throughout the organization, as well as budgeting, analyzing investment proposals, and providing funds for operations.
- **Marketing** is responsible for assessing consumer wants and needs, and selling and promoting the organization's products or services.
- **Operations** is responsible for producing the goods or providing the services offered by the organization.

Supply Chain

What is Supply chain?

- The sequence of organizations, including their **facilities**, **functions**, and **activities** that are involved in producing and delivering a product or service.



Supply Chain

Facilities

- *Warehouses*
- *Factories*
- *Processing centers*
- *Offices*
- *Distribution centers*
- *Retail stores*

Functions and activities

- *Forecasting*
- *Purchasing*
- *Inventory management*
- *Information management*
- *Quality assurance*
- *Scheduling*
- *Production*
- *Distribution*
- *Delivery*
- *Customer service*

Supply Chain

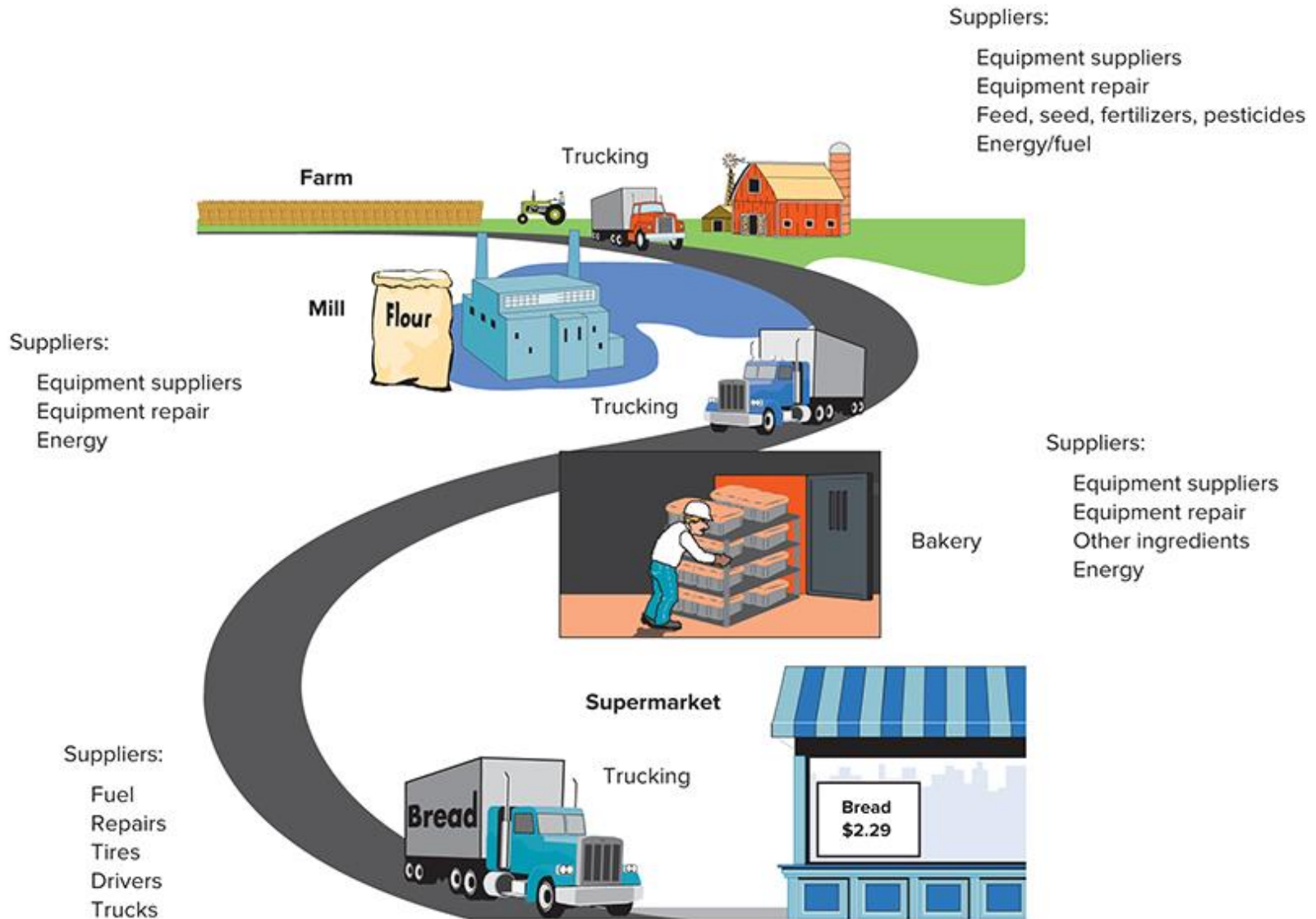
External parts of a supply chain

- *raw materials, parts, equipment, supplies, and/or other inputs* to the organization, and they deliver outputs that are *goods* to the organization's customers.

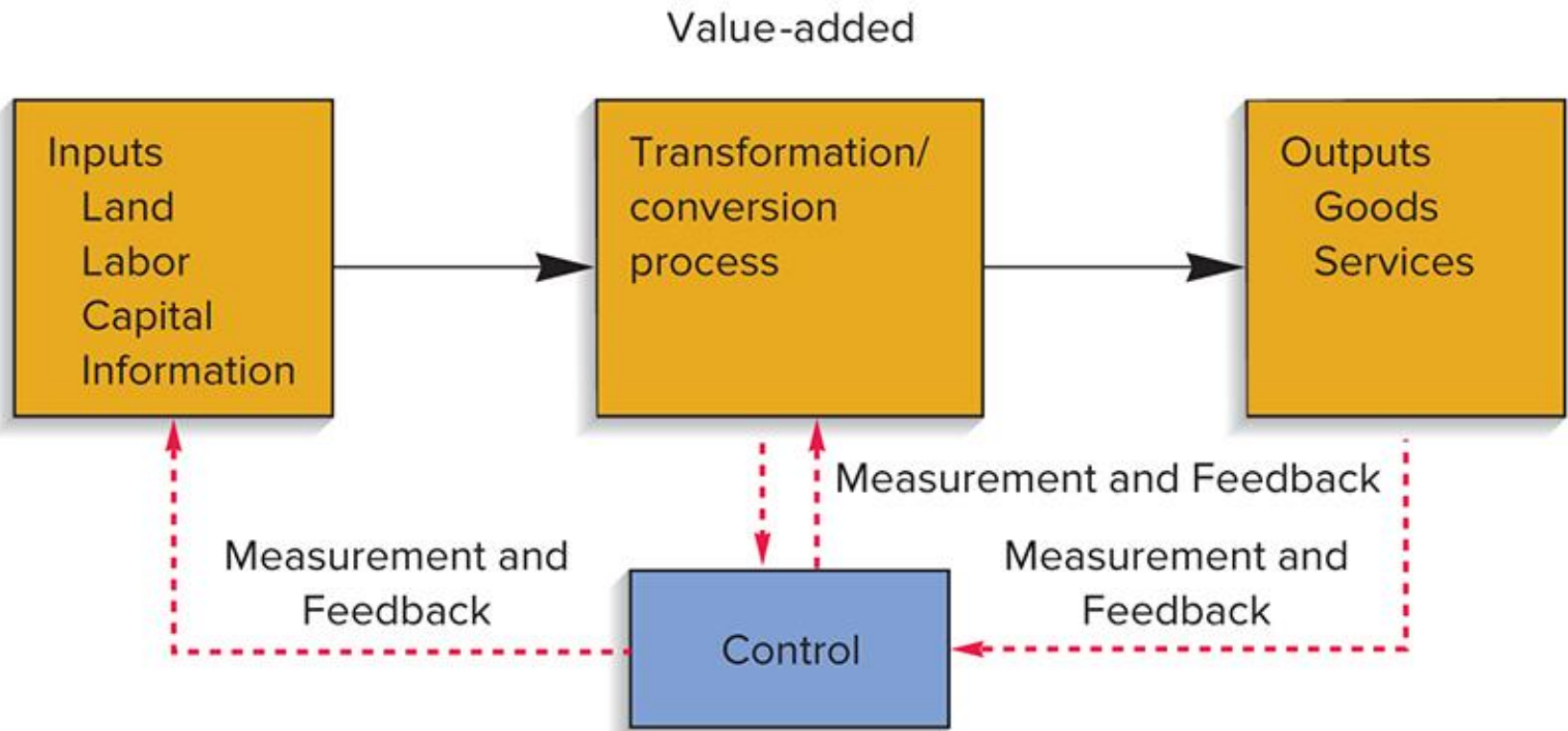
Internal parts of a supply chain

- *operations function* itself, supplying operations with parts and materials, performing work on products, and/or performing services.

Supply Chain



The Transformation Process



Feedback = *Measurements* taken at various points in the transformation process

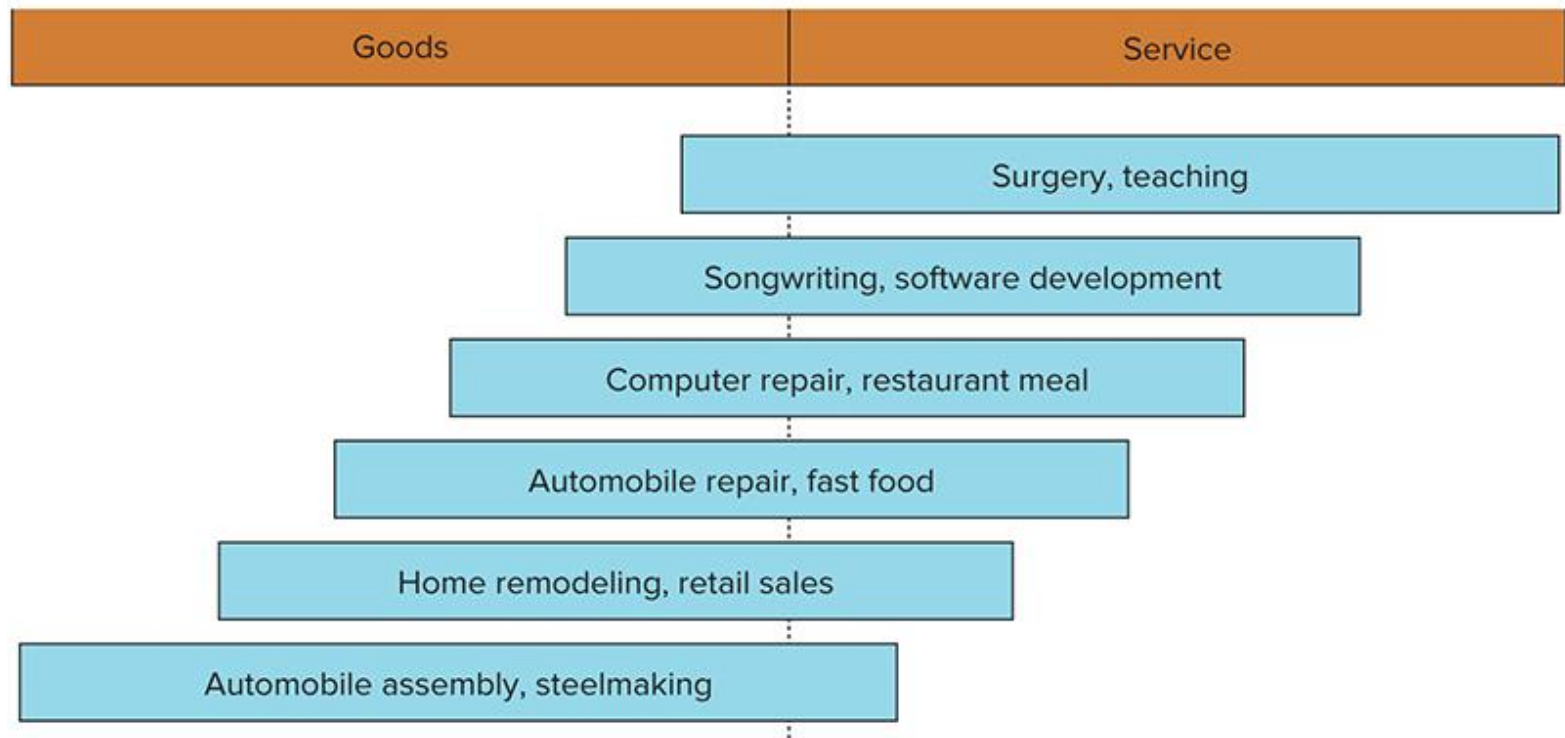
Control = The *comparison* of feedback against previously established standards to determine if corrective action is needed

The Transformation Process

	Inputs	Processing	Output
Food Processor	Raw vegetables	Cleaning	Canned vegetables
	Metal sheets	Making cans	
	Water	Cutting	
	Energy	Cooking	
	Labor	Packing	
	Building	Labeling	
	Equipment		
Hospital	Doctors, nurses	Examination	Treated patients
	Hospital	Surgery	
	Medical supplies	Monitoring	
	Equipment	Medication	
	Laboratories	Therapy	

Goods-service Continuum

Products are typically neither purely service- or purely goods-based



Good vs. Service

Differences in products and services:

1. Degree of customer contact
2. Labor content of jobs
3. Uniformity of input
4. Uniformity of output
5. Measurement of productivity
6. Production and delivery
7. Quality assurance
8. Amount of inventory
9. Evaluation of work
10. Ability to patent design

Good vs. Service

Characteristic	Goods	Services
Output	Tangible	Intangible
Customer contact	Low	High
Labor content	Low	High
Uniformity of input	High	Low
Measurement of productivity	Easy	Difficult
Opportunity to correct problems before delivery	High	Low
Inventory	Much	Little
Wages	Narrow range	Wide range
Patentable	Usually	Not usually

Why Study Operations Management?

Every aspect of business affects or is affected by operations, and many service jobs are closely related to operations

- Financial services
- Marketing services
- Accounting services
- Information services

Through learning about operations and supply chains you will have a better understanding of:

- The world you live in
- The global dependencies of companies and nations
- Reasons that companies succeed or fail
- The importance of working with others

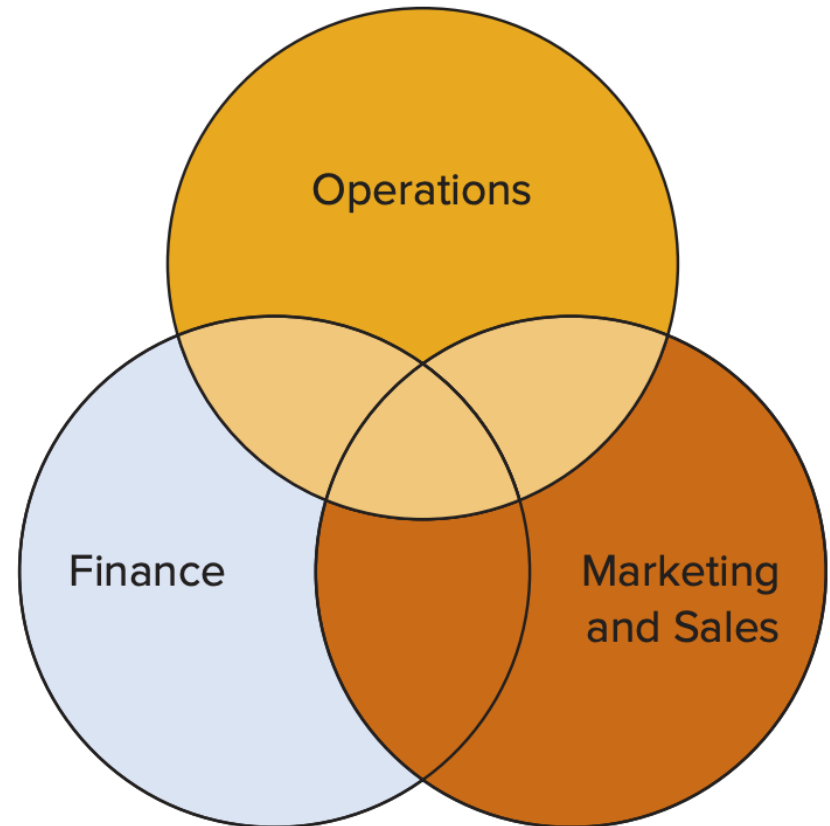
Function Overlap

Finance & operations

- Budgeting
- Economic analysis of investment proposals
- Provision of funds

Marketing & operations

- Demand data
- Product and service design
- Competitor analysis
- Lead time data



OM and Supply Chain Career Opportunities

- Operations manager
- Supply chain manager
- Production analyst
- Schedule coordinator
- Production manager
- Industrial engineer
- Purchasing manager
- Inventory manager
- Quality manager



OM and Supply Chain Career Opportunities

People who work in the *operations* or *supply chain* field should have two important skills:

Communication skills

- Mentoring ability
- Collaboration ability
- Negotiation ability
- Communication ability

Knowledge skills

- Product or service knowledge
- Process knowledge
- Industry and global knowledge
- Financial and accounting skills
- Project management skills

OM-Related Professional Societies

- APICS - The Association for Operations Management
- American Society for Quality (ASQ)
- Institute for Supply Management (ISM)
- Institute for Operations Research and Management Science (INFORMS)
- The Production and Operations Management Society (POMS)
- The Project Management Institute (PMI)
- Council of Supply Chain Management Professionals (CSCMP)



Process Management

Process - one or more actions that transform inputs into outputs

Three Categories of Business Processes:

Upper-management processes	These govern the operation of the entire organization, e.g. governance of enterprise, decision making for operation strategy.
Operational processes	These are core processes that make up the value stream, e.g. purchasing, production of goods or services, marketing, and sales.
Supporting processes	These support the core operational processes, and involve the cooperation of many function areas, e.g. accounting, marketing, human resources, and IT.

Supply & Demand

Operations &
Supply Chains

Sales & Marketing

Supply $>$ Demand

Wasteful
Costly

Supply $<$ Demand

Opportunity Loss
Customer
Dissatisfaction

Supply $=$ Demand

Ideal

Process Variation

Four Sources of Variation:

Variety of goods or services being offered	The greater the <i>variety of goods and services offered</i> , the greater the variation in production or service requirements.
Structural variation in demand	Includes <i>trends</i> and <i>seasonal demand</i> variations. These variations are predictable, and the demand variation has huge influences for capacity planning.
Random variation	The <i>uncontrollable changes in processes and demand</i> . It is typically not be influenced and controlled by managers.
Assignable variation	This type of change has <i>identifiable sources</i> , such as the investments of materials, capacity of machine and labor. it can be corrected through analysis and evaluation actions.

Note: Variations can be disruptive to operations and supply chain processes. They may result in additional costs, delays and shortages, poor quality, and inefficient work systems.

Scope of Operations Management

The scope of operations management ranges across the organization.

The operations function includes many interrelated activities such as:

- Forecasting
- Capacity planning
- Locating facilities
- Facilities and layout
- Scheduling
- Managing inventories
- Assuring quality
- Motivating employees
- And more . . .

Role of the Operations Manager

The Operations function consists of all activities *directly* related to producing goods or providing services.

A primary function of the **operations manager** is to guide the system by decision making.

- **System design decisions**
- **System operation decisions**

System Design Decisions

Affect the performance and operation of organization for many years into the future, including:

- **Capacity**
- **Facility location and layout**
- **Product and service planning**
- **Acquisition and placement of equipment**

Such decisions usually involve the *long-term commitment* of resources, and are high-impact decisions.

Failure to make good design decisions may reduce the performance and competitiveness of the organization in the marketplace.

System Operation Decisions

Generally tactical or operational in general, and making regular and ongoing, including:

- **Management of personnel**
- **Inventory management and control**
- **Scheduling**
- **Project management**
- **Quality assurance**

The decisions essentially determines many of the *parameters of production and operations*, these parameters are directly affected by design decisions. Therefore, operations managers spend more time on these decisions than on any other decision area.

Operations Manager Decision Making

Most operations decisions involve many alternatives that can have quite different impacts on costs or profits. Therefore, typical operations decisions must be included:

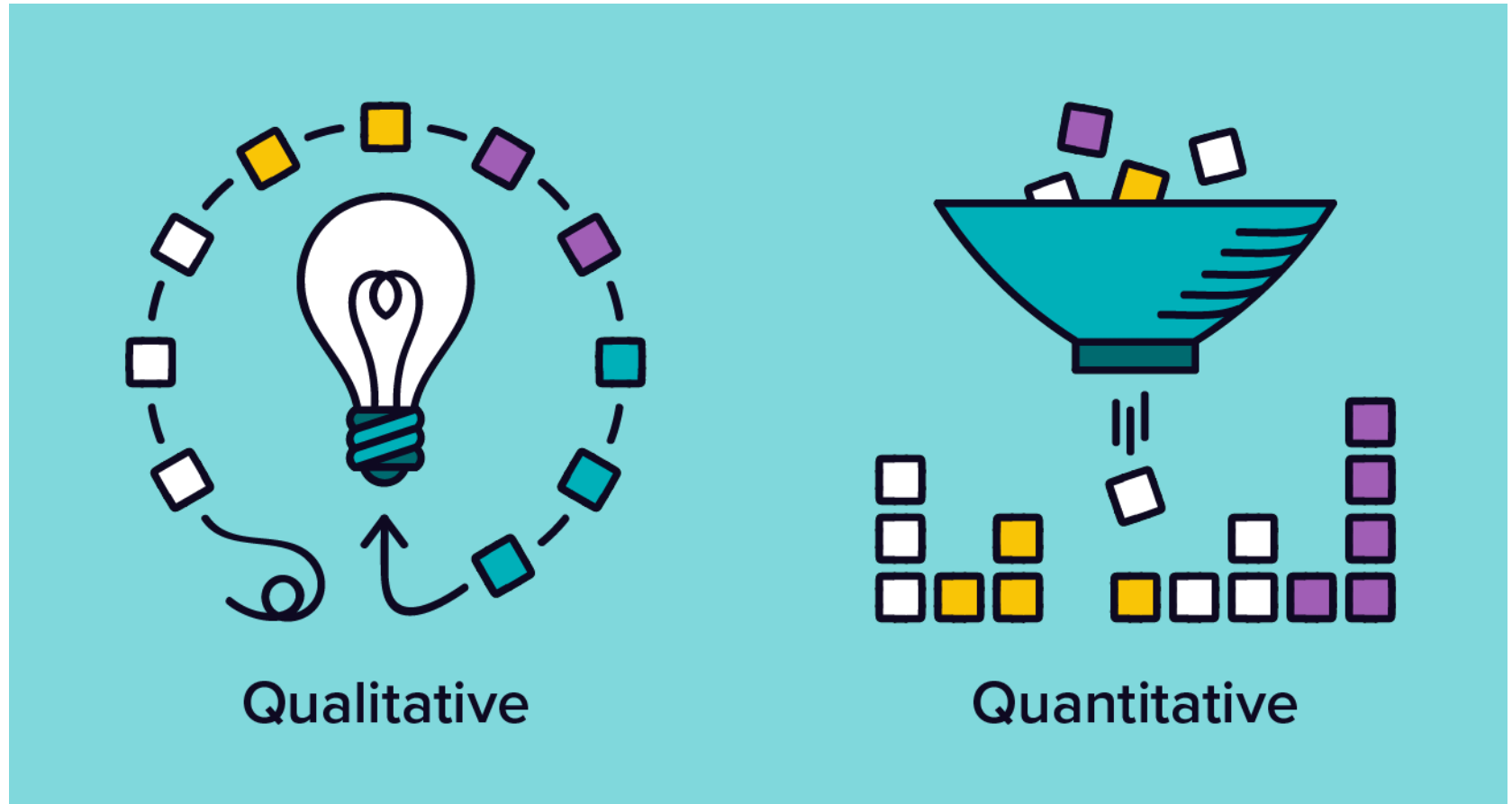
- **What:** What resources are needed, and in what amounts?
- **When:** When will each resource be needed? When should the work be scheduled? When should materials and other supplies be ordered?
- **Where:** Where will the work be done?
- **How:** How will the product or service be designed? How will the work be done? How will resources be allocated?
- **Who:** Who will do the work?

Quantitative Approaches

A decision-making approach that frequently seeks to obtain a *mathematically optimal solution*.

- Widely accepted because of *calculators* and *computers* handling the required calculations.
- Growing availability of *software packages*.
- Managers typically use a combination of *qualitative* and *quantitative* approaches for decision making.

Qualitative V.S Quantitative Approaches



[Quantitative vs. Qualitative Research: The Differences Explained | Scribbr](#) 

Metrics and Trade-Offs

Performance metrics in manage and control operations:

- Profits
- Costs
- Quality
- Productivity
- Flexibility
- Inventories
- Schedules
- Forecast accuracy

Trade-offs decision problem

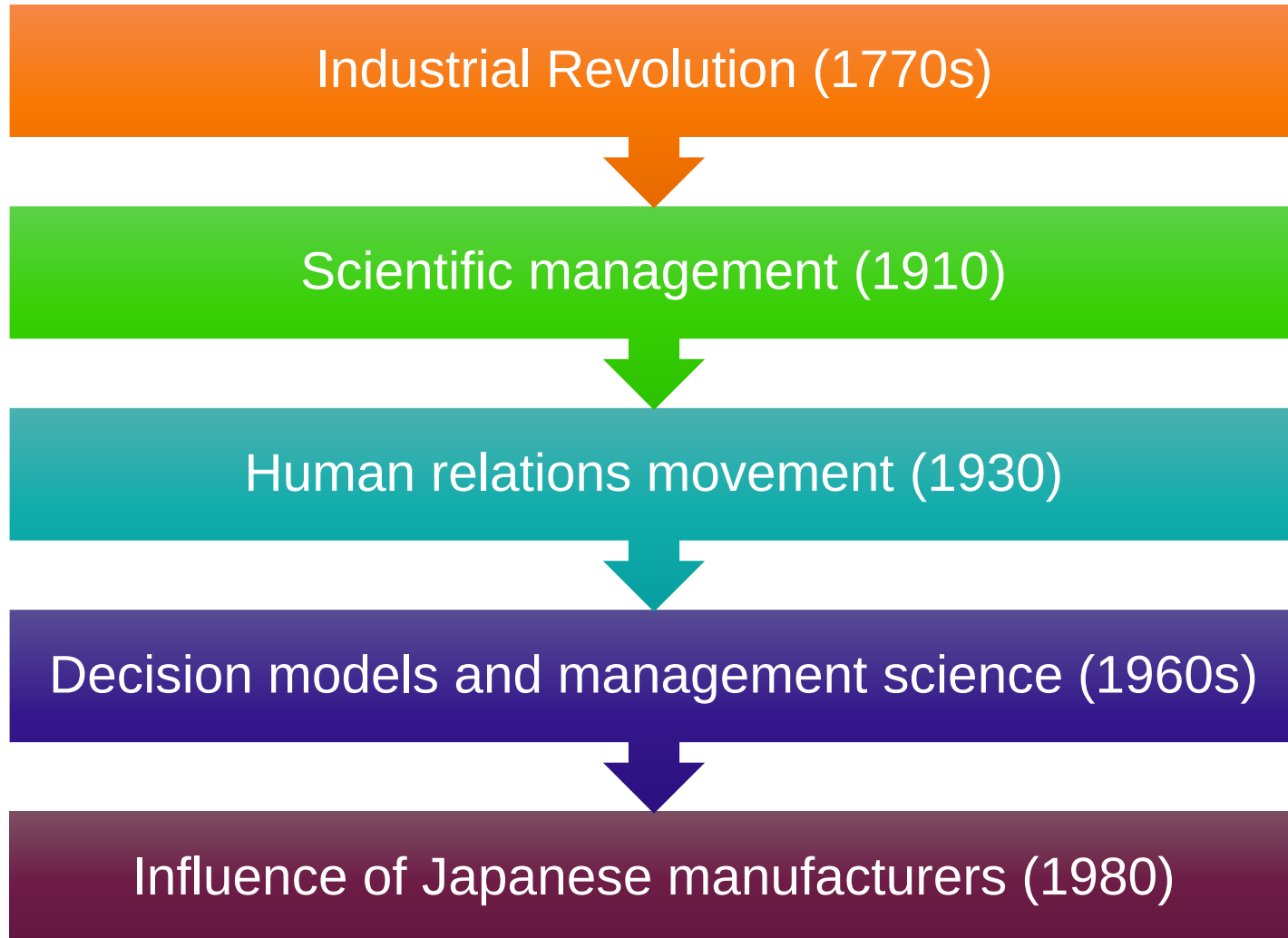
- A trade-off is giving up one thing in return for something else
 - E.g. carrying more inventory (an expense) in order to achieve a greater level of customer service

Degree of Customization

Degree of customization of products or services offered is a major influence on the entire organization

- Highly customized goods or services may be:
 - More labor intensive
 - More time consuming
 - Requires highly skilled labor
 - Requires more flexible equipment
 - Lower volume
 - Higher price tag
 - Affects other departments

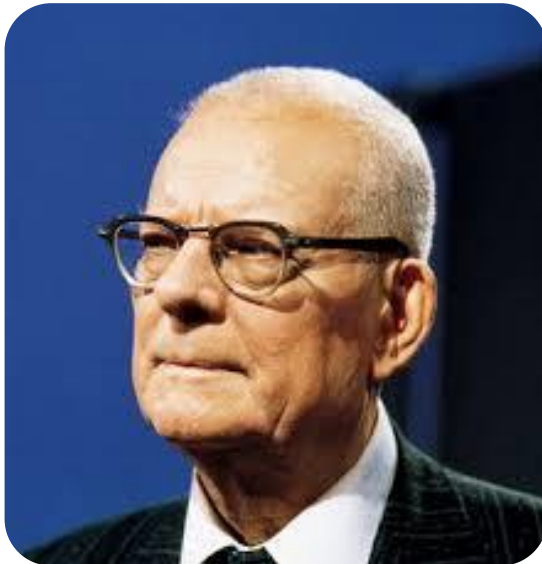
Historical Evolution of OM



Influence of Japanese Manufacturers

Refined and developed management practices that increased productivity.

- Credited with fueling the “quality revolution”
- Proposed Just-in-Time (JIT) production



W Edwards Deming, 1900-1993



Joseph M. Juran, 1904-2008

Historical summary of operations management

Approximate Date	Contribution/Concept	Originator
1776	Division of labor	Adam Smith
1790	Interchangeable parts	Eli Whitney
1911	Principles of scientific management	Frederick W. Taylor
1911	Motion study, use of industrial psychology	Frank and Lillian Gilbreth
1912	Chart for scheduling activities	Henry Gantt
1913	Moving assembly line	Henry Ford
1915	Mathematical model for inventory ordering	F.W. Harris
1930	Hawthorne studies on worker motivation	Elton Mayo
1935	Statistical procedures for sampling and quality control	H.F. Dodge, H. G. Romig, W. Shewhart, L.H.C. Tippet
1940	Operations research applications in warfare	Operations research groups
1947	Linear programming	George Dantzig
1951	Commercial digital computers	Sperry Univac, IBM
1950s	Automation	Numerous
1960s	Extensive development of quantitative tools	Numerous
1960s	Industrial dynamics	Jay Forrester
1975	Emphasis on manufacturing strategy	W. Skinner
1980s	Emphasis on flexibility, time-based competition, lean production	T. Ohno, S. Shingo, Toyota
1980s	Emphasis on quality	W. Edwards Deming, J. Juran, K. Ishikawa
1990s	Internet, supply chain management	Numerous
2000s	Applications service providers and outsourcing Social media, YouTube, and others	Numerous Numerous

LO 1.8

Operations Today

- Technology management
- Global competition
- Working with fewer resources
- Revenue management
- Agility

Reading: Agility Creates a Competitive Edge

- [ZARA's Business Model via Its Operation Strategy](#)
- [Zara's strategy](#)



Key Issues for Operations Managers Today

Economic
conditions of
Market environment

Organization's
innovating ability

Quality problems

Risk management

Maintain the Cyber-
security

Increase
competitiveness in
a global economy

Environmental Concerns

Sustainability

- Using resources in ways that do not harm ecological systems that support human existence.
- Sustainability measures often go beyond traditional environmental and economic measures to include measures that incorporate social criteria in decision making.



Sustainable Development Goals (SDGs)



LO 1.10

Ethical Issues in Operations

Ethical issues that may arise in many aspects of operations management:

- Financial statements
- Worker safety
- Product safety
- Quality
- The environment
- The community
- Hiring and firing workers
- Closing facilities
- Workers' rights

When making decisions, operations managers must consider how their decisions will affect **key stakeholders**, including: *shareholders, employees, customers, the community, even whole environment.*

Reading: Sustainable in Business

- Sustainability at The Hershey Company



The Need for Supply Chain Management

In the past, organizations did little to manage the supply chain beyond their own operations and immediate suppliers which led to numerous problems.

- Oscillating inventory levels
- Inventory stockouts
- Late deliveries
- Quality problems

Both domestic and international competition has intensified, the pressure has mounted to improve the management of organizational supply chains.

Supply Chain Issues

1. The need to improve operations
2. Increasing levels of outsourcing
3. Increasing transportation costs
4. Competitive pressures
5. Increasing globalization
6. Increasing importance of e-business
7. The complexity of supply chains
8. The need to manage inventories
9. The need to deal with trade wars

Element of Supply Chain Management

Key elements of supply chain management:

Element	Typical Issues
Customers	Determining what products and/or services customers want
Forecasting	Predicting the quantity and timing of customer demand
Design	Incorporating customers, wants, manufacturability, and time to market
Capacity planning	Matching supply and demand
Processing	Controlling quality, scheduling work
Inventory	Meeting demand requirements while managing the costs of holding inventory
Purchasing	Evaluating potential suppliers, supporting the needs of operations on purchased goods and services
Suppliers	Monitoring supplier quality, on-time delivery, and flexibility; maintaining supplier relations
Location	Determining the location of facilities
Logistics	Deciding how to best move information and materials

LO 1.11

Summary

- Operations consists of **processes** that convert **inputs** into **outputs**.
- The operations function in business organizations is responsible for **producing products** and **providing services**.
- Operations management involves **system design** and **operating decisions** related to product and/or service.
- Supply chains are the sequential system of **suppliers** and **customers** that begins with basic sources of inputs and ends with final customers of the system.
- **Operations** and **supply chains** are interdependent. One couldn't exist without the other, and no business organization could exist without both.
- Ethical behavior and Sustainability is an important part and future target of good management practice for all enterprises.



The End
